AP Calculus — Topic-by-Topic Practice Set

Units 1–5 (Differentiation)

13 topics × 20 questions = 260 questions in total

Full answer key included · print-ready

How to use this booklet

1. The booklet contains 13 topics (from the source table) with 20 questions each — 260 questions total.
2. The complete answer key is placed at the end ("Answer Key"), so you can check your work afterward.
3. Suggested pacing: 20–30 minutes per topic (20 questions), marking wrong answers in red.
4. After finishing, note which topics produced errors — that is your focus list for the next round.

Questions

1. Types of Discontinuities

Key idea: Removable: the limit exists but differs from f(a) (or f(a) is undefined). Jump: both one-sided limits exist but are unequal. Infinite: one or both sides tend to ±∞. Oscillating: e.g. sin(1/x).

1. What type of discontinuity does $f(x)=(x^2-1)/(x-1)$ have at $x=1$?

2. What type of discontinuity does $f(x)=1/x$ have at $x=0$?

3. $f(x)=\{ x, x<1 ; x+1, x\geq 1 \}$. What type of discontinuity at $x=1$?

4. What type of discontinuity does $f(x)=\sin(1/x)$ have at $x=0$?

5. What type of discontinuity does $f(x)=(x^2-4)/(x-2)$ have at $x=2$?

6. What type of discontinuity does $f(x)=|x|/x$ have at $x=0$?

7. What type of discontinuity does $f(x)=1/(x-2)^2$ have at $x=2$?

8. What type of discontinuity does $f(x)=(x^2-9)/(x+3)$ have at $x=-3$?

9. $f(x)=\{ x-1, x<0 ; x+1, x\geq 0 \}$. What type of discontinuity at $x=0$?

10. What type of discontinuity does $f(x)=\cos(1/x)$ have at $x=0$?

11. What type of discontinuity does $f(x)=(\sin x)/x$ have at $x=0$?

12. What type of discontinuity does $f(x)=\tan x$ have at $x=\pi/2$?

13. What type of discontinuity does $f(x)=\lfloor x \rfloor$ (floor function) have at $x=2$?

14. What type of discontinuity does $f(x)=1/\sqrt{x}$ have at $x=0$ (from the right)?

15. What type of discontinuity does $f(x)=(x^3-1)/(x-1)$ have at $x=1$?

16. $f(x)=\{ x^2, x<2 ; 5, x=2 ; 2x+1, x>2 \}$. What type of discontinuity at $x=2$?

17. What type of discontinuity does $f(x)=\ln x$ have at $x=0$ (from the right)?

18. What type of discontinuity does $f(x)=(x-3)/(x^2-9)$ have at $x=3$?

19. What type of discontinuity does f(x)=e^(1/x) have at $x=0$?

20. $f(x)=\{ x+2, x<1 ; x^2+1, x\geq 1 \}$. What type of discontinuity at $x=1$?

2. Intermediate Value Theorem

Key idea: If f is continuous on the closed interval [a,b], then f takes every value between f(a) and f(b). Commonly used to prove that an equation has a solution.

1. f is continuous on $[0,2]$, $f(0)=-1$, $f(2)=3$. Must there exist c with $f(c)=1$?

2. Does $f(x)=x^3-x-1$ have a zero on $[1,2]$?

3. Does $f(x)=\cos x$ attain the value 0.5 on $[0,\pi]$?

4. f is continuous on $[1,3]$, $f(1)=2$, $f(3)=8$. Must there exist c with $f(c)=5$?

5. Does $f(x)=x^2-2$ have a zero on $[0,2]$?

6. f is continuous on $[-1,1]$, $f(-1)=-3$, $f(1)=3$. Must there exist c with $f(c)=0$?

7. Does f(x)=e^x−3 have a zero on $[1,2]$?

8. Does $f(x)=x^3+2x-1$ have a zero on $[0,1]$?

9. What hypothesis is required to apply the Intermediate Value Theorem?

10. f is continuous on $[0,1]$, $f(0)=0$, $f(1)=1$. Must there exist c with $f(c)=1/2$?

11. Does $f(x)=\ln x-1$ have a zero on $[2,3]$?

12. For $f(x)=x^2-5x+6$ on $[0,1]$, does the IVT guarantee a zero?

13. f is continuous on $[2,6]$, $f(2)=-4$, $f(6)=4$. Must there exist c with $f(c)=0$?

14. Does $f(x)=x^4-3$ have a zero on $[1,2]$?

15. f is continuous on $[0,4]$, $f(0)=10$, $f(4)=2$. Must there exist c with $f(c)=6$?

16. Does f(x)=2^x−5 have a zero on [2,3]?

17. f is continuous on $[-2,2]$ with values -1 and 1. Must its graph cross 0?

18. Does f(x)=sin x−0.7 have a zero on $[0,\pi/2]$?

19. f is continuous on $[1,5]$, $f(1)=3$, $f(5)=3$. Must there exist c with $f(c)=5$?

20. f is continuous on $[0,3]$, $f(0)=-2$, $f(3)=7$. Which value is NOT guaranteed to be attained: 0 / 5 / 10?

3. Horizontal Asymptotes

Key idea: If $\lim(x\to+\infty) f(x)=L$ or $\lim(x\to-\infty) f(x)=M$, then $y=L$ or $y=M$ is a horizontal asymptote; the two sides may differ.

1. Find the horizontal asymptote of $f(x)=(3x^2+1)/(x^2-1)$.

2. Find the horizontal asymptote of $f(x)=1/x$.

3. Find the horizontal asymptote of $f(x)=(2x+1)/(x-3)$.

4. Find the horizontal asymptote of $f(x)=5x^3/(x^3+2)$.

5. Find the horizontal asymptote of f(x)=e^(−x) as $x\to+\infty$.

6. Find the horizontal asymptote(s) of f(x)=arctan x.

7. Find the horizontal asymptote of $f(x)=(x^2-1)/(2x^2+3)$.

8. Find the horizontal asymptote of $f(x)=4x/(x^2+1)$.

9. Find the horizontal asymptote of $f(x)=(x^3+1)/(2x^3-x)$.

10. Find the horizontal asymptote of $f(x)=3$.

11. Find the horizontal asymptote of $f(x)=(2x^2+3x)/x^2$.

12. Find the horizontal asymptote of f(x)=(sin x)/x as $x\to+\infty$.

13. Find the horizontal asymptote of $f(x)=(1-2x^2)/(3x^2+4)$.

14. Find the horizontal asymptote(s) of $f(x)=x/\sqrt{(x^2+1)}$.

15. Find the horizontal asymptote of f(x)=e^x/(e^x+1) as $x\to+\infty$.

16. Find the horizontal asymptote of $f(x)=(6x^2-1)/(3x^2+x)$.

17. Find the horizontal asymptote of f(x)=ln x/x as $x\to+\infty$.

18. Find the horizontal asymptote of $f(x)=(x+2)/(x^2-1)$.

19. Find the horizontal asymptote of f(x)=2^x/(2^x+3) as $x\to+\infty$.

20. Find the horizontal asymptote of $f(x)=5x/(x+1)$.

4. Removable Discontinuities

Key idea: If $\lim(x\to a) f(x)$ exists but differs from f(a) (or f(a) is undefined), simply define f(a) to equal that limit to make f continuous at a.

1. $f(x)=(x^2-1)/(x-1)$. What value of f(1) makes f continuous at x=1?

2. $f(x)=(x^2-4)/(x-2)$. What value of f(2) makes f continuous?

3. $f(x)=(x^2-9)/(x-3)$. What value of f(3) makes f continuous?

4. f(x)=(sin x)/x. What value of f(0) makes f continuous?
5. f(x)=(1−cos x)/x. What value of f(0) makes f continuous?
6. f(x)=(x³−8)/(x−2). What value of f(2) makes f continuous?
7. f(x)=(√x−1)/(x−1). What value of f(1) makes f continuous?
8. f(x)=(x²+x−6)/(x−2). What value of f(2) makes f continuous?
9. f(x)=(x²−5x+6)/(x−2). What value of f(2) makes f continuous?
10. f(x)=(e^x−1)/x. What value of f(0) makes f continuous?
11. f(x)=(x³−1)/(x−1). What value of f(1) makes f continuous?
12. f(x)=(x⁴−1)/(x−1). What value of f(1) makes f continuous?
13. f(x)=(√(x+3)−2)/(x−1). What value of f(1) makes f continuous?
14. f(x)=(tan x)/x. What value of f(0) makes f continuous?
15. f(x)=(x²−16)/(x+4). What value of f(−4) makes f continuous?
16. f(x)=(2x²−3x)/x. What value of f(0) makes f continuous?
17. f(x)=(1−√x)/(1−x). What value of f(1) makes f continuous?
18. lim(h→0)[(x+h)²−x²]/h = ? (treat x as a constant)
19. f(x)=(x²−2x)/x. What value of f(0) makes f continuous?
20. f(x)=(x³+x²)/x². What value of f(0) makes f continuous?

5. Product Rule

Key idea: (uv)' = u'v + uv'.

1. y=x²·sin x, find y'.
2. y=x·e^x, find y'.
3. y=(x²+1)(x−3), find y'.
4. y=x³·ln x, find y'.
5. y=(2x+1)(x²−1), find y'.
6. y=x²·e^x, find y'.
7. y=sin x·cos x, find y'.
8. y=(x+1)(x+2), find y'.
9. y=x·cos x, find y'.
10. y=(x²+3)(2x−5), find y'.
11. y=√x·(x+1), find y'.
12. y=x·ln x, find y'.
13. y=(3x²+1)(x−2), find y'.
14. y=e^x·sin x, find y'.
15. y=x²(x³+1), find y'.
16. y=(x²−1)(x²+1), find y'.
17. y=x³·e^x, find y'.
18. y=(2x+3)(x²−x), find y'.
19. y=(x+2)cos x, find y'.
20. y=x²·√x, find y'.

6. Definition of the Derivative

Key idea: $f'(x) = \lim_{h\to 0}[f(x+h)-f(x)]/h = \lim_{x\to a}[f(x)-f(a)]/(x-a)$.

1. Use the definition to find the derivative of $f(x)=x^2$ at $x=3$.
2. Use the definition to find the derivative of $f(x)=3x+1$.
3. Use the definition to find the derivative of $f(x)=x^3$.
4. Use the definition to find the derivative of $f(x)=\sqrt{x}$.
5. Use the definition to find the derivative of $f(x)=1/x$.
6. Use the definition to find the derivative of $f(x)=5$.
7. Use the definition to find the derivative of $f(x)=x^2+2x$.
8. Use the definition to find the derivative of $f(x)=2x^2$.
9. $\lim_{h\to 0}[(2+h)^2-4]/h$ = ? (this is the derivative of some function at $x=2$)
10. $\lim_{h\to 0}[\sqrt{(9+h)}-3]/h$ = ?
11. Use the definition to find the derivative of $f(x)=x^2-3x$.
12. $\lim_{x\to 2}(x^2-4)/(x-2)$ = ? (the derivative of x^2 at $x=2$)
13. Use the definition to find the derivative of $f(x)=1/(x+1)$.
14. $\lim_{h\to 0}[(3+h)^3-27]/h$ = ?
15. Use the definition to find the derivative of $f(x)=x^4$.
16. Use the definition to find the derivative of $f(x)=\sqrt{(x+1)}$.
17. $\lim_{h\to 0}\sin(h)/h$ = ?
18. Find the derivative of $f(x)=x^2$ at $x=5$.
19. $\lim_{x\to 1}(x^3-1)/(x-1)$ = ?
20. Use the definition to find the derivative of $f(x)=1/x^2$.

7. Average Rate of Change of Trigonometric Functions

Key idea: Average rate of change = $[f(b)-f(a)]/(b-a)$. Answers often contain π.

1. Find the average rate of change of $f(x)=\sin x$ on $[0, \pi/2]$.
2. Find the average rate of change of $f(x)=\cos x$ on $[0, \pi]$.
3. Find the average rate of change of $f(x)=\sin x$ on $[0, \pi]$.
4. Find the average rate of change of $f(x)=\tan x$ on $[0, \pi/4]$.
5. Find the average rate of change of $f(x)=\cos x$ on $[0, \pi/2]$.
6. Find the average rate of change of $f(x)=\sin x$ on $[0, \pi/6]$.
7. Find the average rate of change of $f(x)=\sin(2x)$ on $[0, \pi/4]$.
8. Find the average rate of change of $f(x)=\cos x$ on $[0, \pi/3]$.
9. Find the average rate of change of $f(x)=\sin x$ on $[\pi/2, \pi]$.
10. Find the average rate of change of $f(x)=\tan x$ on $[0, \pi/6]$.
11. Find the average rate of change of $f(x)=\cos(2x)$ on $[0, \pi/2]$.
12. Find the average rate of change of $f(x)=\sin x$ on $[0, \pi/3]$.
13. Find the average rate of change of $f(x)=\cos x$ on $[\pi/3, \pi/2]$.
14. Find the average rate of change of $f(x)=\sin x$ on $[\pi/3, \pi/2]$.
15. Find the average rate of change of $f(x)=\sin x$ on $[0, 2\pi]$.

16. Find the average rate of change of f(x)=cos x on [π/2, π].
17. Find the average rate of change of f(x)=sin x on [π/4, π/2].
18. Find the average rate of change of f(x)=tan x on [π/4, π/3].
19. Find the average rate of change of f(x)=cos x on [0, 2π].
20. Find the average rate of change of f(x)=sin x on [π/6, π/2].

8. Derivative of arcsin (Difference Quotient)

Key idea: (arcsin x)' = 1/√(1−x²) for −1<x<1. It can be proved with the difference-quotient limit lim(h→0) [arcsin(x+h)−arcsin x]/h.

1. d/dx (arcsin x) = ?
2. Find (arcsin x)' at x=0.
3. Find (arcsin x)' at x=1/2.
4. Find (arcsin x)' at x=√2/2.
5. lim(h→0)[arcsin(h)−arcsin(0)]/h = ?
6. d/dx arcsin(2x) = ?
7. d/dx arcsin(x²) = ?
8. Find (arcsin x)' at x=√3/2.
9. d/dx [x·arcsin x] = ?
10. d/dx arcsin(x/2) = ?
11. lim(h→0)[arcsin(1/2+h)−arcsin(1/2)]/h = ?
12. d/dx [(arcsin x)²] = ?
13. d/dx arcsin(e^x) = ?
14. Find (arcsin x)' at x=−1/2.
15. d/dx [arcsin x + arccos x] = ?
16. d/dx arcsin(3x) = ?
17. Give lim(h→0)[arcsin(x+h)−arcsin x]/h in terms of x.
18. Find (arcsin x)' at x=0.6.
19. d/dx arcsin(√x) = ?
20. lim(x→0) arcsin(x)/x = ?

9. Product + Chain Rule

Key idea: (uv)' = u'v + uv'; [f(g(x))]' = f'(g(x))·g'(x). The two rules are often used together.

1. y=x²·sin(3x), find y'.
2. y=x·e^(2x), find y'.
3. y=(x²+1)³, find y'.
4. y=x³·cos(x²), find y'.
5. y=x·sin(x²), find y'.
6. y=(2x+1)⁵, find y'.
7. y=x·e^(x²), find y'.
8. y=x·ln(x²+1), find y'.

9. $y=x\cdot\sqrt{(x^2+1)}$, find y'.
10. $y=x^2\cdot\cos(3x)$, find y'.
11. $y=(x^2-1)^4$, find y'.
12. $y=x^2\cdot e^{(-x)}$, find y'.
13. $y=\sin x\cdot\cos(2x)$, find y'.
14. $y=x\cdot\sin(x^3)$, find y'.
15. $y=(3x^2+2)^4$, find y'.
16. $y=x\cdot e^{(\sin x)}$, find y'.
17. $y=x^2\cdot\ln(3x)$, find y'.
18. $y=x\cdot\tan(x^2)$, find y'.
19. $y=x\cdot\sqrt{(\sin x)}$, find y'.
20. $y=(x+1)^2\cdot e^x$, find y'.

10. Rectilinear Motion

Key idea: Position s(t); velocity v=s'(t); acceleration a=v'(t)=s''(t). The particle is instantaneously at rest when v=0 and moves in the positive direction when v>0.

1. $s(t)=t^3-6t^2+9t$, find v(t).
2. $s(t)=t^3-6t^2+9t$, find a(t).
3. $s(t)=t^3-6t^2+9t$. At what times is the particle at rest?
4. $s(t)=t^3-6t^2+9t$. When does the particle move in the positive direction?
5. $s(t)=t^3-6t^2+9t$, find s(0).
6. $s(t)=2t^2+3t$, find v(t).
7. $s(t)=2t^2+3t$, find a(t).
8. $s(t)=t^2-4t$. When is the particle at rest?
9. $s(t)=5t-t^2$, find v(t).
10. $s(t)=5t-t^2$. When is the position a maximum?
11. $s(t)=3t^3$, find v(t).
12. $s(t)=3t^3$, find a(t).
13. $s(t)=t^3-3t$, find v(t).
14. $s(t)=t^3-3t$ ($t\geq0$). When is the particle at rest?
15. $s(t)=16t^2$, find v(t).
16. $s(t)=16t^2$, find a(t).
17. $s(t)=t^2-6t+5$. Find the velocity at t=2.
18. $s(t)=t^2-6t+5$. Find the acceleration.
19. $s(t)=t^3-6t^2+9t$. Find the velocity at t=2 and state the direction of motion.
20. $s(t)=t^3-6t^2+9t$. Find the displacement from t=0 to t=4.

11. Ladder-Related Rates

Key idea: Let the ladder have length L, with its base x from the wall and its top at height y, so $x^2+y^2=L^2$. Differentiate with respect to t: $2x\cdot dx/dt + 2y\cdot dy/dt = 0$, i.e. $dy/dt = -(x/y)\cdot dx/dt$.

1. A 10 ft ladder has its base 6 ft from the wall and slides away at 2 ft/s. Find the rate at which the top slides down.

2. A 13 ft ladder has its base 5 ft from the wall, sliding away at 3 ft/s. Find the speed of the top.

3. A 5 m ladder has its base 3 m from the wall, sliding away at 0.5 m/s. Find the speed of the top.

4. A 10 ft ladder has its base 8 ft from the wall, sliding away at 1 ft/s. Find the speed of the top.

5. A 15 ft ladder has its base 9 ft from the wall, sliding away at 2 ft/s. Find the speed of the top.

6. A 20 ft ladder has its base 12 ft from the wall, sliding away at 1.5 ft/s. Find the speed of the top.

7. A 10 ft ladder has its base 6 ft from the wall, sliding away at 2 ft/s. Find the speed of the top (as a fraction).

8. For a 10 ft ladder, at what base distance does the top descend at the same rate at which the base moves away?

9. A 13 ft ladder has its base 12 ft from the wall, sliding away at 2 ft/s. Find the speed of the top.

10. A 10 ft ladder has its base 6 ft from the wall and its top 8 ft high; the base slides away at 2 ft/s. Find the rate of change of the triangle's area.

11. A 5 m ladder has its base 3 m from the wall and moves toward the wall at 0.5 m/s. Find the speed of the top.

12. A 10 ft ladder has its base 6 ft from the wall, sliding away at 2 ft/s. Find the downward speed of the top.

13. A 25 ft ladder has its base 7 ft from the wall, sliding away at 1 ft/s. Find the speed of the top.

14. A 25 ft ladder has its base 15 ft from the wall, sliding away at 3 ft/s. Find the speed of the top.

15. As the base of a sliding ladder approaches the wall ($x \to L$), what happens to the downward speed of the top?

16. A 12 ft ladder has its base 6 ft from the wall, sliding away at 2 ft/s. Find the speed of the top.

17. A 17 ft ladder has its base 8 ft from the wall, sliding away at 4 ft/s. Find the speed of the top.

18. A 10 ft ladder has its base 8 ft from the wall and moves toward the wall at 1 ft/s. Find the speed of the top.

19. A 13 ft ladder has its base 5 ft from the wall, sliding away at 1.3 ft/s. Find the speed of the top.

20. Write the general relation linking dx/dt and dy/dt in the ladder problem.

12. Related Rates

Key idea: Differentiate the geometric or physical relation (area, volume, distance, …) with respect to t, then substitute the instantaneous values. Common formulas: $A=\pi r^2$, $V=(4/3)\pi r^3$, $A=s^2$, $V=s^3$.

1. The radius of a circle grows at 2 cm/s. Find the rate of change of the area when r=5 cm.

2. The area of a circle grows at 10π cm²/s. Find dr/dt when r=5 cm.

3. The radius of a sphere grows at 1 cm/s. Find dV/dt when r=3.

4. A balloon's volume grows at 100. Find dr/dt when r=5.

5. A square's side grows at 3. Find dA/dt when s=4.

6. A cube's side grows at 2. Find dV/dt when s=3.

7. The area of a circle grows at 6π. Find dr/dt when r=3.

8. The circumference of a circle grows at 4π. Find dr/dt.

9. Two cars: one travels north at 60 mph and the other east at 80 mph. When they are 3 and 4 miles from the intersection, find the rate of change of the distance between them.

10. The radius of a ripple grows at 3 cm/s. Find the area growth rate when r=10.

11. $y=x^2$ and dx/dt=2. Find dy/dt when x=3.

12. $y=x^3$ and dx/dt=1. Find dy/dt when x=2.

13. $y=\sqrt{x}$ and dx/dt=4. Find dy/dt when x=9.

14. y=1/x and dx/dt=1. Find dy/dt when x=2.

15. A square's area grows at 20. Find ds/dt when s=5.

16. A sphere's volume grows at 4π. Find dr/dt when r=1.

17. A right triangle has legs x=4 and y=3; x grows at 2 per second and y shrinks at 1 per second. Find the rate of change of the area.

18. y=sin x and dx/dt=2. Find dy/dt when x=0.

19. The area of a circle grows at 8π. Find dr/dt when r=2.

20. The surface area of a sphere is $S=4\pi r^2$; r grows at 0.5. Find dS/dt when r=3.

13. Mean Value Theorem

Key idea: If f is continuous on [a,b] and differentiable on (a,b), then there exists $c\in(a,b)$ such that $f'(c) = [f(b)-f(a)]/(b-a)$. First compute the secant slope, then set the derivative equal to it and solve for c.

1. $f(x)=x^2$ on [0,2]. Find c guaranteed by the MVT.

2. $f(x)=x^2$ on [1,3]. Find c.

3. $f(x)=x^3$ on [0,3]. Find c.

4. $f(x)=x^3$ on [0,1]. Find c.

5. $f(x)=x^2+2x$ on [0,1]. Find c.

6. f(x)=1/x on [1,2]. Find c.

7. $f(x)=\sqrt{x}$ on [1,4]. Find c.

8. $f(x)=x^3-x$ on [−1,1]. Find all values of c.

9. $f(x)=x^2$ on [−1,1]. Find c.

10. f(x)=ln x on [1,e]. Find c.

11. f(x)=e^x on [0,1]. Find c (express with ln).

12. $f(x)=x^2-4x$ on [0,4]. Find c.

13. $f(x)=x^3$ on [−1,1]. Find c.

14. f(x)=sin x on $[0,\pi]$. Find c.

15. $f(x)=x^2$ on [0,4]. Find c.

16. f(x)=x+1/x on [1,2]. Find c.

17. $f(x)=\cos x$ on $[0,\pi/2]$. Find c (express with arcsin).

18. $f(x)=x^2-x$ on $[0,2]$. Find c.

19. $f(x)=x^4$ on $[0,2]$. Find c.

20. $f(x)=2x^2-3x$ on $[1,3]$. Find c.

Answer Key

1. Types of Discontinuities

1. Removable (lim = 2, but f(1) is undefined)
2. Infinite
3. Jump (left limit 1, right limit 2)
4. Oscillating (the limit does not exist)
5. Removable (lim = 4)
6. Jump (left limit -1, right limit 1)
7. Infinite
8. Removable (lim = -6)
9. Jump (left limit -1, right limit 1)
10. Oscillating
11. Removable (lim = 1)
12. Infinite
13. Jump (left limit 1, right limit 2)
14. Infinite ($\to +\infty$ as $x \to 0^+$)
15. Removable (lim = 3)
16. Jump (left limit 4, right limit 5)
17. Infinite ($\to -\infty$ as $x \to 0^+$)
18. Removable (lim = 1/6)
19. Infinite (right limit $+\infty$, left limit 0; an essential discontinuity)
20. Jump (left limit 3, right limit 2)

2. Intermediate Value Theorem

1. Yes (1 lies between -1 and 3)
2. Yes ($f(1)=-1<0$, $f(2)=5>0$)
3. Yes ($\cos 0=1$, $\cos \pi=-1$; 0.5 lies between)
4. Yes (5 lies between 2 and 8)
5. Yes ($f(0)=-2$, $f(2)=2$)
6. Yes
7. Yes ($f(1)\approx -0.28<0$, $f(2)=e^2-3\approx 4.39>0$)
8. Yes ($f(0)=-1$, $f(1)=2$)
9. f must be continuous on the closed interval [a,b]
10. Yes
11. Yes ($f(2)<0$, $f(3)>0$)
12. No ($f(0)=6>0$ and $f(1)=2>0$, same sign)
13. Yes
14. Yes ($f(1)=-2$, $f(2)=13$)
15. Yes (6 lies between 2 and 10)
16. Yes ($f(2)=-1<0$, $f(3)=3>0$)
17. Yes
18. Yes ($\sin 0=0<0.7$, $\sin(\pi/2)=1>0.7$)
19. No (5 is not between 3 and 3)
20. 10 (outside the range $[-2,7]$)

3. Horizontal Asymptotes

1. $y=3$
2. $y=0$
3. $y=2$
4. $y=5$
5. $y=0$
6. $y=\pi/2$ ($x\to+\infty$) and $y=-\pi/2$ ($x\to-\infty$)
7. $y=1/2$
8. $y=0$
9. $y=1/2$
10. $y=3$ (a constant function is its own asymptote)
11. $y=2$
12. $y=0$
13. $y=-2/3$
14. $y=1$ ($x\to+\infty$) and $y=-1$ ($x\to-\infty$)
15. $y=1$
16. $y=2$
17. $y=0$
18. $y=0$
19. $y=1$
20. $y=5$

4. Removable Discontinuities

1. 2
2. 4
3. 6
4. 1
5. 0
6. 12
7. 1/2
8. 5
9. -1
10. 1
11. 3
12. 4
13. 1/4
14. 1
15. -8
16. -3
17. 1/2
18. $2x$
19. -2
20. 1

5. Product Rule

1. $2x \sin x + x^2 \cos x$

2. e^x(1+x)
3. $3x^2-6x+1$
4. $3x^2 \ln x + x^2$
5. $6x^2+2x-2$
6. $e^x(x^2+2x)$
7. $\cos 2x$
8. $2x+3$
9. $\cos x - x \sin x$
10. $6x^2-10x+6$
11. $(x+1)/(2\sqrt{x}) + \sqrt{x}$
12. $\ln x + 1$
13. $9x^2-12x+1$
14. $e^x(\sin x + \cos x)$
15. $5x^4+2x$
16. $4x^3$
17. $x^2e^x(x+3)$
18. $6x^2+2x-3$
19. $\cos x - (x+2)\sin x$
20. $(5/2)x^{(3/2)}$

6. Definition of the Derivative

1. 6
2. 3
3. $3x^2$
4. $1/(2\sqrt{x})$
5. $-1/x^2$
6. 0
7. $2x+2$
8. $4x$
9. 4
10. 1/6
11. $2x-3$
12. 4
13. $-1/(x+1)^2$
14. 27
15. $4x^3$
16. $1/(2\sqrt{(x+1)})$
17. 1
18. 10
19. 3
20. $-2/x^3$

7. Average Rate of Change of Trigonometric Functions

1. $2/\pi$
2. $-2/\pi$
3. 0

4. $4/\pi$
5. $-2/\pi$
6. $3/\pi$
7. $4/\pi$
8. $-3/(2\pi)$
9. $-2/\pi$
10. $2\sqrt{3}/\pi$
11. $-4/\pi$
12. $3\sqrt{3}/(2\pi)$
13. $-3/\pi$
14. $(6-3\sqrt{3})/\pi$
15. 0
16. $-2/\pi$
17. $(4-2\sqrt{2})/\pi$
18. $12(\sqrt{3}-1)/\pi$
19. 0
20. $3/(2\pi)$

8. Derivative of arcsin (Difference Quotient)

1. $1/\sqrt{(1-x^2)}$
2. 1
3. $2\sqrt{3}/3$ $(=2/\sqrt{3})$
4. $\sqrt{2}$
5. 1
6. $2/\sqrt{(1-4x^2)}$
7. $2x/\sqrt{(1-x^4)}$
8. 2
9. arcsin x + $x/\sqrt{(1-x^2)}$
10. $1/\sqrt{(4-x^2)}$
11. $2/\sqrt{3}$
12. 2 arcsin x $/\sqrt{(1-x^2)}$
13. e^x/√(1−e^(2x))
14. $2\sqrt{3}/3$
15. 0
16. $3/\sqrt{(1-9x^2)}$
17. $1/\sqrt{(1-x^2)}$
18. 5/4 (=1.25)
19. $1/(2\sqrt{(x(1-x))})$
20. 1

9. Product + Chain Rule

1. $2x \sin 3x + 3x^2 \cos 3x$
2. e^(2x)(1+2x)
3. $6x(x^2+1)^2$
4. $3x^2\cos(x^2) - 2x^4\sin(x^2)$
5. $\sin(x^2) + 2x^2\cos(x^2)$

6. $10(2x+1)^4$
7. $e^{(x^2)}(1+2x^2)$
8. $\ln(x^2+1) + 2x^2/(x^2+1)$
9. $(2x^2+1)/\sqrt{(x^2+1)}$
10. $2x \cos 3x - 3x^2 \sin 3x$
11. $8x(x^2-1)^3$
12. $e^{(-x)}(2x-x^2)$
13. $\cos x \cos 2x - 2 \sin x \sin 2x$
14. $\sin(x^3) + 3x^3\cos(x^3)$
15. $24x(3x^2+2)^3$
16. $e^{(\sin x)}(1 + x \cos x)$
17. $2x \ln(3x) + x$
18. $\tan(x^2) + 2x^2\sec^2(x^2)$
19. $\sqrt{(\sin x)} + x \cos x/(2\sqrt{(\sin x)})$
20. $e^{x(x+1)(x+3)}$

10. Rectilinear Motion

1. $v(t)=3t^2-12t+9$
2. $a(t)=6t-12$
3. $t=1$ and $t=3$
4. $t<1$ or $t>3$ (i.e. $0\leq t<1$ and $t>3$)
5. 0
6. $4t+3$
7. 4
8. $t=2$
9. $5-2t$
10. $t=2.5$
11. $9t^2$
12. $18t$
13. $3t^2-3$
14. $t=1$
15. $32t$
16. 32
17. -2
18. 2
19. $v=-3$, moving in the negative direction
20. 4

11. Ladder-Related Rates

1. -1.5 ft/s
2. -1.25 ft/s
3. -0.375 m/s
4. $-4/3$ ft/s
5. -1.5 ft/s
6. -1.125 ft/s
7. $-3/2$ ft/s

8. $x=y=5\sqrt{2}\approx 7.07$ ft
9. -4.8 ft/s
10. 3.5 ft²/s
11. $+0.375$ m/s (rising)
12. 1.5 ft/s (downward)
13. $-7/24$ ft/s
14. -2.25 ft/s
15. It tends to infinity
16. $-2/\sqrt{3}$ ft/s
17. $-32/15$ ft/s
18. $+4/3$ ft/s (rising)
19. $-13/24\approx -0.542$ ft/s
20. $x\cdot dx/dt + y\cdot dy/dt = 0$, i.e. $dy/dt = -(x/y)dx/dt$

12. Related Rates

1. 20π cm²/s
2. 1 cm/s
3. 36π
4. $1/\pi$
5. 24
6. 54
7. 1
8. 2
9. 100 mph
10. 60π cm²/s
11. 12
12. 12
13. 2/3
14. $-1/4$
15. 2
16. 1
17. 1
18. 2
19. 2
20. 12π

13. Mean Value Theorem

1. $c=1$
2. $c=2$
3. $c=\sqrt{3}$
4. $c=1/\sqrt{3}=\sqrt{3}/3$
5. $c=1/2$
6. $c=\sqrt{2}$
7. $c=9/4$
8. $c=\pm 1/\sqrt{3}$
9. $c=0$

10. $c = e - 1$
11. $c = \ln(e - 1)$
12. $c = 3/2$
13. $c = 0$
14. $c = \pi/2$
15. $c = 2$
16. $c = \sqrt{2}$
17. $c = \arcsin(2/\pi)$
18. $c = 1$
19. $c = \sqrt[3]{2} = 2^{(1/3)}$
20. $c = 2$